# PeakCast Wheater Chatbot – Concept (Phase 1)

Markus Pointner - IU14161364, Master Artificial Intelligence

PeakCast is a chatbot for a weather channel that answers forecast questions for hikers, climbers, ski tourers and paragliders in Austria. The channel wants a reliable, low-maintenance service that extends its reach on mobile. The user wants a fast and precise answer on the go, often at the trailhead or take-off site. General descriptions are often not enough for them: wind speed and gusts decide whether a paraglider can launch, cloud cover whether a summit has a view or not and fresh snow matters for a ski tour

The user aks in plain English and get temperature, sky condition (sunny/Cloudy), wind speed and gusts and rain or snowfall for a chosen place and day. Each answer is one or two sentence long with exact values, units and a short outdoor hint, for example “Strong gusts in the afternoon – fly in the morning.” PeakCast furthermore remembers the last place, so follow-ups such as “And what about tomorrow?” work. A conversation is limited to about ten question and answers. It informs the user but it does not replace official mountain weather or avalance reports

Instead of a hosted framework like Rasa or Dialogflow, PeakCast is built in Python, so every component stays transparent, no license key or cloud account is needed and the bot runs after a single install. A scikit-learn classifier trained on example sentences detects the intent (temperature, sky, wind, precipitation, greeting or out of scope). Places are matched against Austrian towns, villages and peaks from the free GeoNames database and day words like “today”, “tomorrow”, “afternoon” or “Sunday” are resolved by rules. The forecast itself comes from the free Open-Meteo API wit a fictional dataset as an offline fallback.

The process below shows one dialogue turn. When the intent is unclear or the place is missing PeakCast asks back instead of guessing. After a goodbye or the tenth answer the conversations ends.

